HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Designed and implemented Python feature pipelines on AWS Lambda and S3 to process ~3,500 events per second, replacing a legacy batch job and improving data freshness for downstream analytics.
- Developed and evaluated a **scikit-learn** gradient-boosted classifier on a ~280k-row transaction dataset, achieving an **AUC of 0.86** for an internal anti-abuse workflow.
- Engineered robust **Java Spring Boot** REST APIs for inter-service communication, incorporating request validation and circuit-breaker patterns to enhance system resilience.
- Authored rigorous, step-by-step **Python** solutions and **SQL** queries to serve as definitive benchmarks for evaluating AI model reasoning and output correctness.
- Critically evaluated AI-generated code outputs using **Scikit-Learn**, identifying and providing structured feedback on technical accuracy, efficiency, and potential data leakage.
- Crafted complex, domain-rich data science challenges for AI models, focusing on areas such as hyperparameter optimization, Bayesian inference, and cross-validation strategies.
- Containerized Python data pipeline services using **Docker** and deployed them to an AWS EKS cluster, configuring basic resource limits for predictable performance and rollout.

SKILLS

Core Data Science – Machine Learning, Statistical Inference, Supervised Learning, Unsupervised Learning, Deep Learning, NLP, Data Engineering

Programming & Scripting – Python, R, SQL

ML Libraries & Frameworks – Scikit-Learn, PyTorch, TensorFlow, pandas, NumPy

Cloud & DevOps – AWS Lambda, S3, Docker, Kubernetes (EKS), CI/CD

Databases & Data Warehousing – PostgreSQL, MySQL, Oracle, SQL Queries

Big Data Technologies – Spark, Hadoop (Concepts)

PROJECTS

AI Reasoning Evaluation Framework

- Developed a Python-based framework to systematically evaluate AI model outputs, focusing on logical consistency, data handling, and adherence to statistical principles.
- Crafted a suite of challenging test cases, including scenarios with imbalanced datasets and potential for overfitting, to probe AI reasoning and identify blind spots.
- Authored detailed ground-truth solutions using **Python** and **SQL**, serving as the definitive benchmark for comparing against AI-generated responses.

Transaction Anomaly Detection Model

- Engineered a feature pipeline in **Python** and **SQL** to extract behavioral signals from a transaction log of approximately 1.1 million records.
- Trained a **LightGBM classifier** on the extracted features, tuning class weights to address a 1:140 fraud imbalance and achieving an **AUC of 0.91** and **F1 score of 0.46** on a held-out month.
- Packaged the trained model behind a **FastAPI** inference endpoint, implementing input validation and logging p95 latency at approximately 180ms.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)